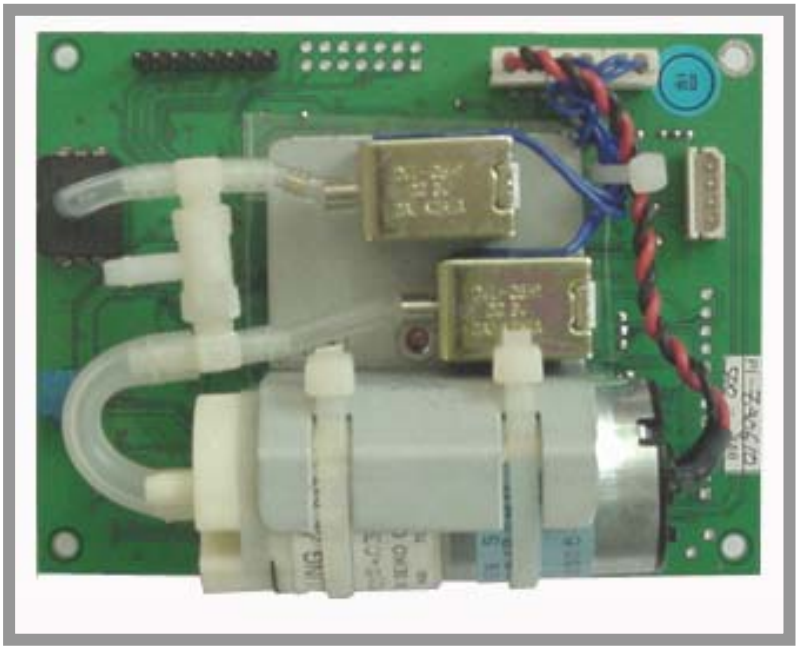
MEKICS *Intensive Care System*

User's Manual

MM300plus



NIBP Module

MEK
Intensive Care System

1. Specification

Technical Specification	Method	*Oscillometric Method (Step Deflation)*
	Pressure	*Measurement Range 0~300mmHg* *Accuracy ±2mmHg or ±2%*
	Cuff	*Reusable Adult Cuff (Standard Accessory)* *Reusable Child Cuff, Disposable Neonate Cuff (optional)*
	Auto Deflation Pressure	*Adult 300mmHg* *Neonate 150mmHg*
	Shut Down Pressure	*330mmHg*
Output	*RS-232C Baud Rate 19200BPS, no partity, 8 Bit, 1 Stop bit standard* *Systolic, Diastolic, Mean Pressure Result, Pressure, Event Message*	
Input	*Start, Stop, Setting Parameter, Debugging Instruction*	
Power	*12V DC 10%, Ripple & Nosie 5% max*	
Power Consumption	*7.2W Type, ±10%, 600mA / 12V Type at Inflation*	
Dimensions	*W102 × D78 × H40 (mm)*	

2. Description of Model Name (MM300p-xx)

Model name	Description
MM300P-S	std RS232 (+/-12V signal direct PC interface), Input Voltage : +/-12V
MM300P-T	TTL level com port (H: +3.3V, L: 0V), Input Voltage : +/-12V

3. Communication protocol

3-1. H/W protocol

RS-232C, 19200 bps, 8 bits, 1 stop bit, No parity, CTS/RTS not used

3-2. S/W protocol

1) Serial Data structure

Normal Type Data structure .

<HEADER> <DATA…> <CRC> <END>

HEADER = 80H ~ EFH (= 80H + 00H~6FH)

DATA & CRC = 00H ~ 7FH

END = F0H ~ FEH (= 80H + 70H~7FH)

* FFH is not used

** MSB=1 of HEADER and END , MSB=0 of DATA thus Data are 7bits .

2) Table of packets

Header Code	Packet Name	Data Length	Direction
60H	Status	5	Tx
62H	Command	? (>= 1)	Rx
64H	Result	6	Tx

▶ [60H] Header of status

Packet structure

```
<HEADER=80H+60H=E0H> <PRSH> <PRSL> <STAT> <EVENT> <FLAG> <CRC>
<END    =80H+7EH=FEH>
```

Bitmap

```
11100000 0ppppppp 0ppppppp 0sssssss 0eeeeeee 0ggggggg 0ccccccc 11111110
  E  0         < PRS >        <STAT>   <EVENT>   <FLAG>     <CRC>    F   E
```

<PRS> ; (pppp~pp) 14 bit Pressure of Cuff [mmHg] Range 0 ~ 400mmHg

<STAT> ; (sss~ss) Status Massage.

0 = DISABLE, 1 = ENABLE, 2 = SEARCH, 3 = MEASURE

<EVENT> Warning=00H~3FH and Error =40H~5FH

** It is possible Warning is occurred in measuring .

** If Error is occurring and the module is Stop and Disable status.

** The module in Error status is initialized by 'RESET' Command.

```
01H = WARN_CUFFHOSE   : CUFF or Hose is abnormal status.
02H = WARN_NOPULSE    : No pulse or Weak pulse.
03H = WARN_PATMOTION  : Patient moving (abnormal Pulse)
04H = WARN_ENVELOPE   : Result of Measuring is  not available
41H = ERR_CUFFHOSE    : Fail for CUFF or hose.
                           ( Fail in Pressing or deflation )
42H = ERR_NOPULSE     : No Pulse
43H = ERR_PATMOTION   : Patient moving
44H = ERR_ENVELOPE    : Fail ; The result of measuring is not available.
51H = ERR_EXCEEDTIME  : Time exceed
52H = ERR_EXCEEDPRES  : Over pressure protection
```

<FLAG> Information for the Module operation status.

```
 40H : H/W SHUTDOWN status .
** It is necessary Power-Reset that The module in H/W Shut-down
   is initialized.
```

20H : Enable status , It is Disabled in Error.
And It is enabled by ' Reset command '
10H : Neonate Mode status.
1:Neonate, 0:Adult
08H : Measuring is OK .
It is cleared by 'RESET, ACK-RESULT or MEASURE' command.
04H : reserved (for internal use)
02H : VALVE 1 (large)
01H : VALVE 2 (small)

▶ [62H] Command

<HEADER=80H+62H=E2H> <CMD> <PARAM> <CRC> <END=80H+7EH=FEH>

11100010 0mmmmmmm . . . 0ccccccc 11111110

E 2 < CMD > < CMD >

<CMD> Command Code

11H = RESET, 21H = READ

RESET : <80H+62H=E2H> <11H> <MODE> <CRC> <80H+7EH=FEH>

<MODE> 0=ADULT, 1=NEONATE

READ : <80H+62H=E2H> <21H> <CRC> <80H+7EH=FEH>

▶ [64H] Result

Packet structure

<HEADER=80H+64H=E4H> <SysH+DiaH><SysL><DiaL><MeanH+PrH><MeanL> <PrL> <CRC>
<END=80H+7EH=FEH>

11100100 00sssddd 0sssssss 0ddddddd 00mmmppp 0mmmmmmm 0ppppppp 0ccccccc

<SysH+DiaH> <SysL> <DiaL> <MeanH+PrH> <MeanL> <PrL> <CRC>

11111110

<Sys> Systolic Pressure

<Dia> Diastolic Pressure

<Mean> Mean Pressure

<Pr> Pulse Rate

3) CRC

7 bit, MSB shifted first, Polynomial = x7 + x6 + x4 + x + 1 .

=Code ==========

```
int CComCalcCRC7(int nCRC, int nSize, PUCHAR pucSrc)
{
    static unsigned char const theTableCRC7[128] =
        {
        0x00,0x53,0x75,0x26,0x39,0x6A,0x4C,0x1F,
       0x72,0x21,0x07,0x54,0x4B,0x18,0x3E,0x6D,
       0x37,0x64,0x42,0x11,0x0E,0x5D,0x7B,0x28,
       0x45,0x16,0x30,0x63,0x7C,0x2F,0x09,0x5A,
        0x6E,0x3D,0x1B,0x48,0x57,0x04,0x22,0x71,
        0x1C,0x4F,0x69,0x3A,0x25,0x76,0x50,0x03,
       0x59,0x0A,0x2C,0x7F,0x60,0x33,0x15,0x46,
       0x2B,0x78,0x5E,0x0D,0x12,0x41,0x67,0x34,
       0x0F,0x5C,0x7A,0x29,0x36,0x65,0x43,0x10,
       0x7D,0x2E,0x08,0x5B,0x44,0x17,0x31,0x62,
       0x38,0x6B,0x4D,0x1E,0x01,0x52,0x74,0x27,
       0x4A,0x19,0x3F,0x6C,0x73,0x20,0x06,0x55,
       0x61,0x32,0x14,0x47,0x58,0x0B,0x2D,0x7E,
       0x13,0x40,0x66,0x35,0x2A,0x79,0x5F,0x0C,
       0x56,0x05,0x23,0x70,0x6F,0x3C,0x1A,0x49,
       0x24,0x77,0x51,0x02,0x1D,0x4E,0x68,0x3B
       };

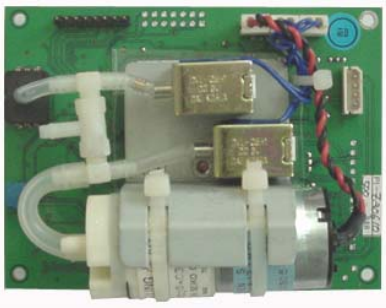
    nCRC &= 0x7F;
    while ( nSize-- > 0 )
    {
nCRC = theTableCRC7[nCRC^((*pucSrc++)&0x7F)];
    }

    return nCRC;
}
```

========== end of code ==========

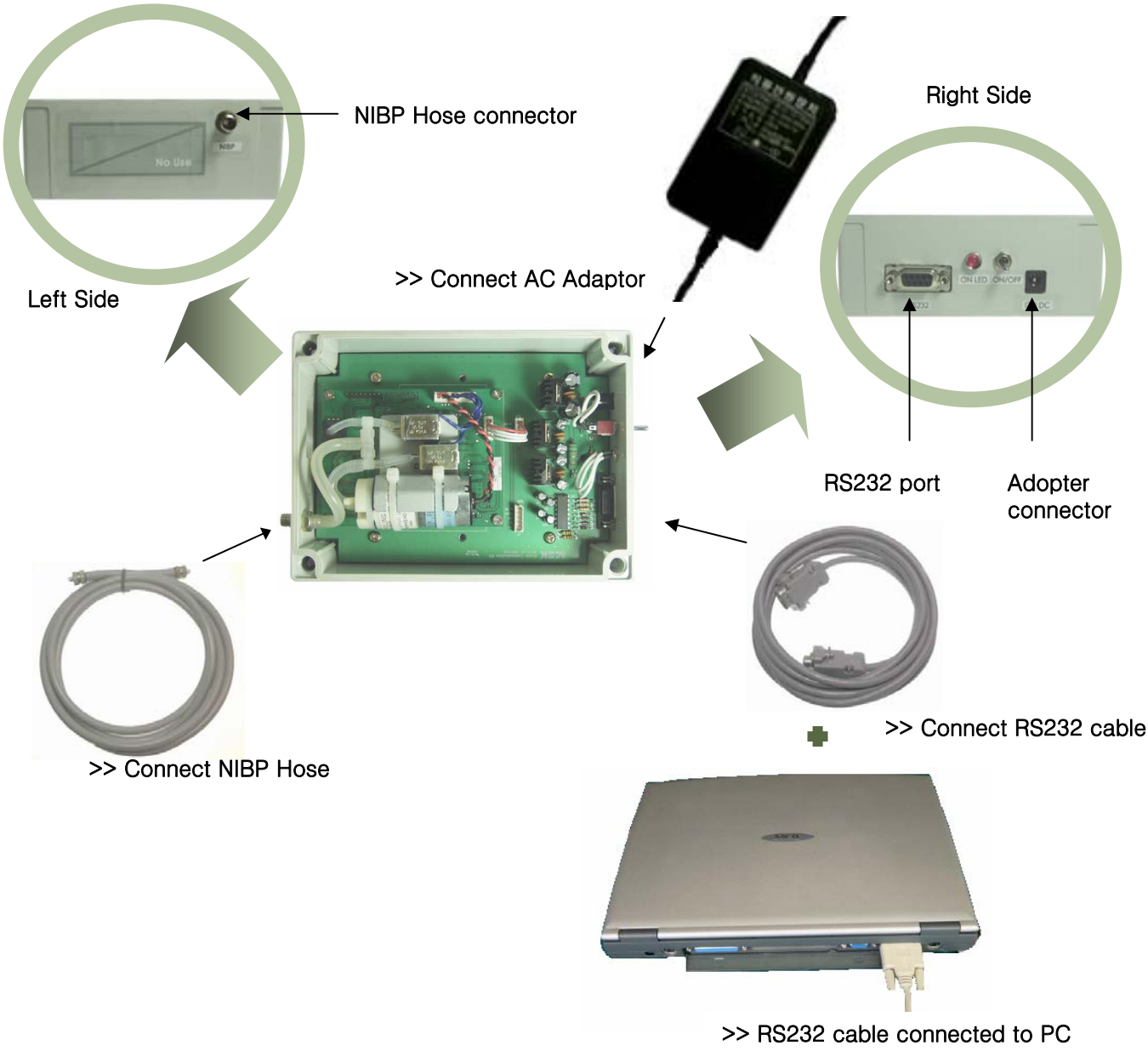
4. List of MM300plus Test Tool Kit (Optional)



NO	Part	Description / Shape
1	MM300 Plus module (Installed)	 >> Front Side  >> Back Side
2	Communication Jig (Installed)	 → AC Adapter → Power → RS232 port

NO	Part	Description / Shape
3	NIBP Adult Cuff	
4	NIBP Hose	
5	RS-232 Cable (Connect PC & Jig)	
6	AC Adapter (220VAC) ** 12V / 1A / 50Hz	
7	PC test Software	3.5″ diskette or Provided by On-line
8	Technical manual	Printed or Provided by On-line

5. Test Kit Assembling



>> When you finish the assembling, please connect NIBP Cuff (part No.5) to NIBP Hose (part No. 3)

>> Plug in Adaptor and then make power on in Test Tool Kit

7. How to Use Test Program 'ViewerNIBP'

1) Complete the Test Tool Kit Assembling
2) Turn on the power of Test Tool Kit (Power Key in right side)
(Make sure the light is on in LED when turn on the power)
3) Double click "ViewerNIBP.exe" test program in your PC
4) Select COM port(COM1 or COM2) which you want, then click " Connect"
5) If the program is working collect, you should see Status Massage
If not, please check the Test Kit Assembling again and reset-up.
6) Connect NIBP Cuff to NIBP hose and then put on Cuff to patient's left Arm or NIBP Simulator if you have a NIBP simulator.
7) To start test, please put "Read" or put Function Key 4 (F4) in your pc keyboard.
8) You can see result of BP(SYS/DIA/MEAN) in program view. Previous result in ( )
9) To restart, you can ①put Reset (F3) and then put "Read(F4)" or ②put "Read(F4)" only without putting "Reset (F3)" (If you do not put Reset, NIBP program calculate proper pressure limit refer to previous result and that make measurement time shorter)
10) When you measure NIBP of Neonate, please check "Neonate(F5)"



>> ViewerNIBP Test Program

No.	Description
①	Select Com port (COM1 or COM2)
②	Start communication with Module
③	Reset condition measurement
④	Start NIBP measurement
⑤	Neonate selectable
⑥	Status of module condition
⑦	Result of present NIBP
⑧	List of Result
⑨	Close Testing

Contact Us

▶ **R&D Department.**

e-mail: delph@mek-ics.com

Tel: 82-2-403-0099 (Ext. 636)

Name: Suk-kyun Hong / Research Engineer

▶ **Module Sales Department.**

e-mail: jisoo@mek-ics.com

Tel: 82-2-403-0099 (Ext. 641)

Name: Jisoo Kim /Sales Manager